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ANAPHORA ANALYSIS USING ACTIVITY VALUE PROPAGATION METHO WITH RAMIFICATION NETWORK

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Abstract

In this paper, we propose the method of performing an anaphora analysis from a semantic structure using Activity Value Propagation Method with Ramification Network. Ramification Network treats semantic structures and anaphoric relations together, and the context can be expressed by one network. In this network, a node corresponds to a word, a concept and an entity. And a link corresponds to a relation of nodes. All nodes and links have activity values. They express the height of the concern and the strength of the connection. We perform correspondence analysis by propagation of activity values. We developed the system on the computer and carried out the experiment.

Key Words: Anaphora, Ramification Network, Activity Value Propagation Method

1. Introduction

In language understanding, context analysis is indispensable. There is an important point which analyzes existence of an antecedent and the candidate for a correspondence to a word in context analysis. Correspondence analysis is required in order to realize a dialog and a high quality machine translation systems [1]. On the other hand, in case that correspondence analysis is performed, the problem that the search range of an antecedent must be determined the existence beforehand, in the conventional method [2][3]. That is, in the method, you have to presume the range in which an antecedent exists beforehand. Therefore, an antecedent cannot be found out when the antecedent appears before the range. Moreover, you have to perform the presumption each a sentence. The method treats all the words in a sentence uniformly without taking a focus into consideration.

We propose the method of performing correspondence analysis of an anaphora from a semantic structure using the Ramification Network. In this paper, since Ramification Network treats semantic structures and anaphoric relations together, the context can be expressed by one network. In the network of this paper, a notation, a concept, and an entity are generated to correspond with a node. And the relation between nodes is generated to correspond with a link. A semantic structure and the context are expressed by connecting a node using the link. All the nodes and links in this network have activity values. The activity value of a node expresses the height of the concern about the word which the node expresses. The activity value of a link expresses the strength of the connection.

In order to perform correspondence analysis, the activity value of the node which expresses the word for analysis target is generated high. Next, in consideration of the influence by the connected link, the activity value of the node is spread to other nodes. Propagation of an activity value is repeated until it will not change. Consequently, the node to which the activity value became high brings an analysis result. Here, by connecting in classification the node which expresses a concept using a thesaurus [4], it is analyzable in consideration of the similarity of a concept. We call it Activity Value Propagation Method.

In this paper, in order to express a case structure simply, we used the network which branched the link. We call this network Ramification Network. This network makes it possible to connect the element expressing case structure directly, and can express case structure clearly. Moreover, the analysis result in the Takeuchi's paper [5] can be used as an input.

The reason using a network is that it can express semantic structures and the context as one network. Moreover, it can analyze without considering restrictions of deciding the range of analysis beforehand, by the above-mentioned method. At this time, the anaphora was set as the target of analysis, because it was going to prove the validity of this paper. However, we think wanting to analyze by improving this paper also about cataphora in the future.

2. Structure of a network

In Ramification Network, there are nodes and links (Figure 1). A node corresponds to the notation of a word, the concept of a word, the concept of a case, and

an entity in context. Four kinds of nodes exist (Table 1). A link usually connects between two nodes, and expresses those relations. For example, the link which connects two word conceptual nodes expresses the vertical relation of concepts. The link which connects a word conceptual node and a notation node matches a word concept and a notation of a word. In Ramification Network, a case structure is expressed by connecting three nodes using the branched link. Three nodes are two word conceptual nodes and a case conceptual node. We call the link Ramification Link.
This network is distinguishable in three layers with the kind of node (Table 2). Conceptual layer expresses concepts and the relation between concepts, such as a vertical relation and a case structure. Notation layer gives a notation to a concept of a conceptual layer and an entity of a context layer. All nodes and links have activity values in this network. And correspondence analysis is performed using these activity values.

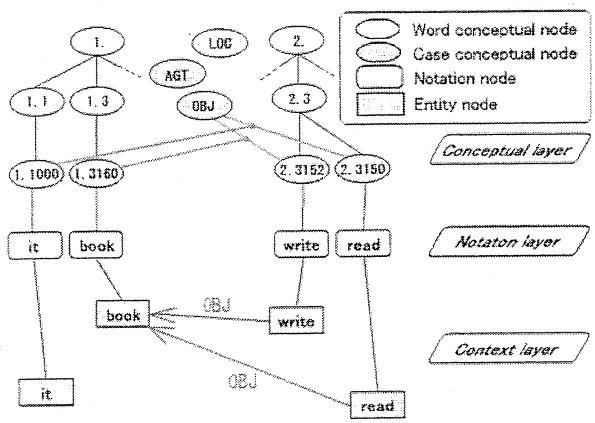


Figure 1: Ramification Network

Table 1: Kinds of node

Node Type	Correspondence to a node
Word conceptual node	The node corresponding to the concept of a word
Case conceptual node	The node corresponding to the concept of a case
Notation node	The node corresponding to the notation of a word
Entity node	The node corresponding to the entity of the word in the context

Table 2: Kinds of layer

Layer Type	Content of Layer
Conceptual layer	The layer containing word conceptual nodes and case conceptual node
Notation layer	The layer containing notation nodes
Context layer	The layer containing entity nodes

2.1. Node

Four kinds of nodes exist. word conceptual node, case conceptual node, notation node, and entity node. A word conceptual node expresses a concept of a word. This node has one conceptual number given by the thesaurus. This number is unique. We describe this node like "1.3160" by the conceptual number in this paper. A case conceptual node expresses a concept of a case. We express a concept of a case using the deep case of Fillmore [6] (Table 3). We describe this node like "OBJ" using abbreviation in this paper. A notation node expresses a notation of a word. This node is connected with a conceptual node and an entity node by a link, and gives the notation to those nodes. We describe this node like "book" using the name in this paper. An entity node expresses an entity in a context. Existence of this node corresponds to an entity of a word in a context. Therefore, only one entity node is generated per existence in a context. If the antecedent of a word "W1" is analyzed with another word "W2, an entity node is not generated every two words of them, however existence of two words are expressed with one entity node. This node doesn't have unique information ('name' and so on). This node is connected with a word conceptual node and a notation node by a link, and obtains a concept and a notation. We describe this node like "she" using name of the entity.

Table 3: Case of Fillmore

Case	Abbreviation
Agent	AGT
Object	OBJ
Experience	EXP
Implement	IMP
Result	RES
Source	SRC
Goal	GOL
Counter Agent	CAG
Time	TIM

2.2. Link

A link usually connects between two nodes, and expresses those relations. We call the link Normal Link. Four kinds of this link exist. The link which connects two word conceptual nodes expresses the vertical relation of a concepts. The vertical relation is understood from two conceptual numbers of the word conceptual nodes. The link which connects a notation node and a word conceptual node matches a notation with the concept. The link which connects a notation node and an entity node matches a notation with the entity. Since an entity node expresses an entity of a word in a context, the link which connects two entity nodes expresses a context. We describe the link which connects node "A" and node "B" like "{A-B}".
In Ramification Network,
In Ramification Network, a case structure is expressed by using the branched link. This link connects three nodes, a word conceptual node of a verb, a case conceptual node, and a word conceptual node. We call this link Ramification Link. We describe the link which connects a word conceptual node of a verb "V", a case conceptual node of a case "C", and a word conceptual node of a word "W" like "{{V-C}-W}".

2.3. Expression of Case Structure

We express a case structure using Ramification Link. First, we connect a conceptual node of a verb "V" and a case conceptual node of a case "C". Next, we branch this link and connect with a word conceptual node of a word "W". We describe this link like "{{V-C}-W}".
This link can connect three nodes composing a case structure directly. Therefore, we can understand a case

structure intuitively. By registering much case structure into a network using this link, we can guess that the verb "V" tends to take the case "C". Figure 2 expresses a case structure "{{"read" - "OBJ"} - "book"}" in Ramification Network.

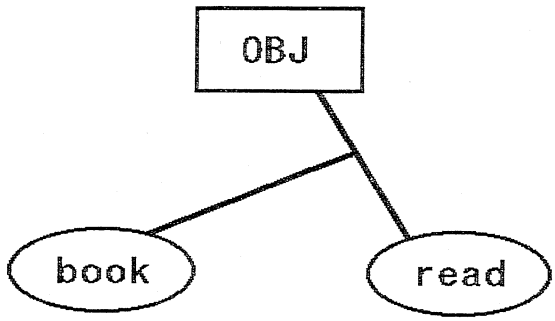


Figure 2: A case structure in Ramification Network

2.4. Expression of Context

We express the context by connecting two entity nodes using a link. An entity node corresponds to an entity in a context. Therefore, even if the same notation appears twice in a context, if they express the same existence, we will generate one entity node only. We express the context to which the entity nodes "E1" and "E2" are connected by a case "C" with the same form as expression of a case structure shown by 2.3, except for the difference between a word conceptual node and an entity node. However, when there are two entity nodes in an anaphoric relation, we express these to one entity node collectively. Figure 3 expresses the result which analyzed two case structures containing the word in an anaphoric relation.

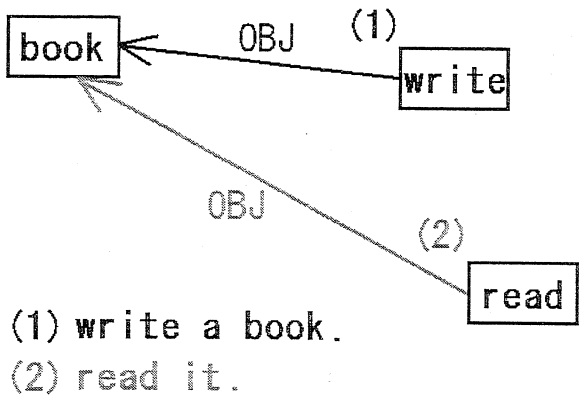


Figure 3: Anaphoric relation

3. Algorithm of Correspondence Analysis

3.1. Activity Value

The activity value of a node expresses the strength of the concern about the word which the node expresses. On the other hand, it is thought that a person has the tendency which gets interested strongly by things newer than things old. Then, such a tendency is imitated and the activity value of a node is decreased with progress of time. The equation 1 means that an activity value decreases with progress of time "t".

$$V(t) = \alpha V(t-1) \quad (\alpha < 1) \qquad (1)$$

"t" increases, whenever a new sentence is inputted. We call "α" the rate of decrease of an activity value. "α" was set to 0.9 this time.

However, it will be stronger interested with the word of an antecedent. Then, the activity value of such a node is increased. The equation 2 means that the activity value of the node increase.

$$V(t) = \beta V(t-1) \quad (\beta > 1) \qquad (2)$$

We call "β" the rate of increase of an activity value.

On the other hand, the activity value of a link expresses the strength of combination between the nodes which it connects.

3.2. Restricted Conditions of Activity Value

An activity value is adjusted to a fixed value in order to complete propagation of an activity value certainly. The equation 3 expresses that the sum of activity value "V(t)" in time t is fixed.

$$\sum V(t) = Const \qquad (3)$$

3.3. Propagation of Activity Value

We make the activity value of a node spread, and perform correspondence analysis. First, the activity value of the node "N1" of the word to analyze is generated high. Next, an adjoining node is generated to spread the activity value of a node "N1" through the link connected to the node "N1". Here, it restricts making an activity value spread directly between close nodes. It is for suppressing the increase in the amount of calculation. However, we will acquire the same effect as the case where an activity value is generated to spread widely, since an activity value is transmitted eventually. The equation 4 expresses that the activity value "Vx(t)" of the node "X" in time "t" is the sum of the activity value of the node "X" and close node "Y" in time "t-1".

$$V_X(t) = V_X(t-1) + \sum W_{XY} V_Y(t-1) \qquad (4)$$

"Wxy" is the joint intensity between "X" and "Y". At this time, "Wxy" took for 1.0. "Y" is the activity value of the node which adjoins "X". As a result of making an activity value spread, the node to which the activity value became the highest is considered as the result of analysis. Here, we set up the conditions for judging convergence of activity values (equation 5).

$$\left| Vx(t) - Vx(t-1) \right| \le \lambda \qquad (5)$$

The equation 5 expresses that the difference of the activity value at Time t and t-1 is below a fixed value "λ". And if the equation 5 is materialized in all nodes, we judge convergence of propagation of activity values. On the other hand, we have to take into consideration the case where the antecedent does not exist. We think that the antecedent does not exist, when there is no node

with the activity value exceeding the value "γ". We call "γ" the value for the antecedent existence judging.

3.4. Update Network

We register the entity node and the link showing case structure into the network based on the result of 3.3. We register the entity node according to the following rules. If the antecedent of the word "W" exists, we will not register the entity node newly. But, if the antecedent does not exist, we will register the entity node newly. After registering the entity node, the system update all activity values.

4. Experiment

We developed the experimental system based on our method on Windows 98 (TM) using C++. This system outputs the activity value of each node in a certain time in a text. We read the texts in each time and know the situation of propagation of an activity value.

4.1. Data

We used 22 sentences (90 case structures, 181 characters) as an input sentence from Japanese-translation sentences of the English teaching-material "Hanasaka Ji-san" [7]. Since the used English teaching-material is easy content and the sentences are written plainly, we used them as the input data.

4.2. Initial state

We made the activity values of all nodes (links) equal in the initial state of this system.

Table 4: Kinds of evaluation

Evaluation	The analysis result of the system
Correct analysis	No antecedent
Incorrect analysis	Incorrect antecedent

In the analysis of the word in which an antecedent does not exist

Evaluation	The analysis result of the system
Correct analysis	Correct antecedent
Incorrect analysis	Incorrect antecedent
No analysis	No antecedent

In the analysis of the word in which an antecedent exists

4.3. Results

We evaluated the experimental results as mentioned below. We distinguished and evaluated the word in which the antecedent exists, and not exists (Table 4). We show the experimental results in Table 5.

4.4. Consideration

We found the following things from this experiment. If the value for the antecedent existence judging "γ" is decreased, a possibility that it is correctly analyzable in the word in which an antecedent exists will increase. On the other hand, the case where this system mistakes in the analysis of the word in which an antecedent does not exist increases.

Next, if the rate of increase of an activity value "β" is increased a focus will be emphasized more. That is, the activity value of a word which becomes an antecedent repeatedly is maintained more highly. and the rate of

Table 5: Results

γ :the value for the antecedent existence judging

β :the rate of increase of an activity value

γ =30 β =1.2

	Correct	Incorrect	No analysis	Total
A	15	22	-----	37
B	0	11	16	27
Total	15	33	16	64

γ =20 β =1.2

	Correct	Incorrect	No analysis	Total
A	11	26	-----	37
B	2	16	9	27
Total	13	42	9	64

γ =20 β =1.5

	Correct	Incorrect	No analysis	Total
A	9	28	-----	37
B	3	18	6	27
Total	12	46	6	64

γ =10 β =1.2

	Correct	Incorrect	No analysis	Total
A	3	34	-----	37
B	2	24	1	27
Total	5	58	1	64

γ =10 β =1.5

	Correct	Incorrect	No analysis	Total
A	3	34	-----	37
B	3	23	1	27
Total	6	57	1	64

correct analysis increases.

5. Experimental System for Parameter Adjustment

We discovered the case where it was not completed by activity values in the experiment of Chapter 4. We consider that this problem is produced for the fault of propagation method of activity values, and a convergence judging. In order to solve this problem, we have to investigate activity values of each node in each time. In the experimental system of Chapter 4, we have to read the activity value of each node in the text which the system outputted. The text is the list of the activity values of each node in each time, and investigating the situation of propagation of activity values by progress of time from these requires time very much.
Then, we developed the new experimental system which can know the situation of propagation of activity values visually and dynamically (Figure 4). This system expresses a part of network of the experimental system of Chapter 4. A square corresponds to a node in Figure 4. In a square, an upper line expresses a name of the node (Figure 5). When this name is a number, the node is a word conceptual node and the number is a conceptual number. When this name is a capital letter, the node is a case conceptual node. And when this name is a small letter, the node is an entity node. A notation node is abbreviated in this system. Instead, an entity node has a name. In a square, a lower numerical value expresses an activity value of the node. And a right-hand side bar expresses the size of the activity value of the node with a bar graph. A line which connects squares in Figure 4 corresponds to a link. The numerical value of the center of a line expresses the weight of the link.

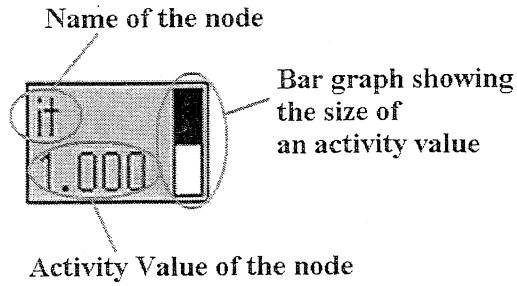


Figure 5: Explanation of the node which the experimental system displayed

This system calculates propagation of activity values with progress of time. We give an activity value to a node in this system, and start this system. Then, we can know the situation that activity values spread with progress of time, by the numerical value and the right-hand side bar graph in a square. Since we can check the situation of propagation of activity values dynamically and visually, we can perform easily improvement of the propagation method of activity values, and adjustment of parameters.
We are developing this system now. We are going to experiment in improvement of the propagation method of activity values and adjustment of parameters from now on using this system.

6. Conclusion

We proposed the method of performing correspondence analysis of an anaphora from a semantic structure using the Ramification Network in this paper. In Ramification Network, four kinds of nodes and links which connect

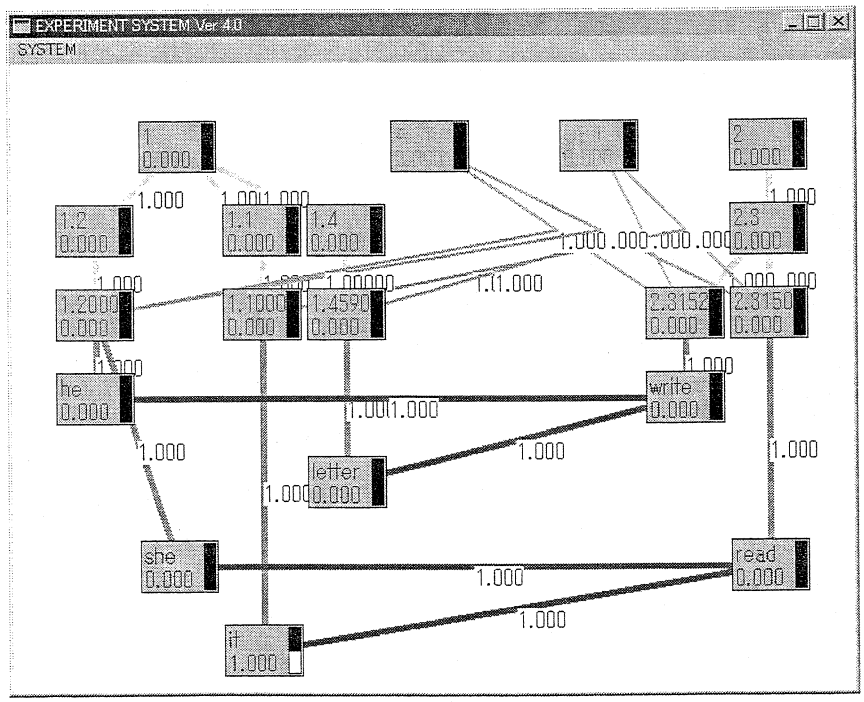


Figure 4: Experimental System for Parameter Adjustment

nodes. A node corresponds to the notation of a word, the concept of a word, the concept of a case, and an entity in context. A link connects between nodes and expresses the vertical relation of concepts and so on. By using these, this network expresses concept, notation, and the context.

We perform correspondence analysis using this network. All the nodes in this network have an activity value. An activity value expresses the degree of the strength of the concern about the node. We give a high activity value to the node of the object of analysis, and make it spread through links. Consequently, we judge that the node with a high activity value is the antecedent.

We experimented by developing a system. Consequently, although the rate of correct analysis was low, by trying the method of patching an activity value and so on showed that it was improvable. As a future subject, we consider how to patch without making an activity value emit.

7. Acknowledgement

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